higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 2

1 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 True
1.2 True
1.3 False
1.4 False
1.5 True

(5 × 1) **[5]**

QUESTION 2

2.1 Chip screening
2.2 Black liquor
2.3 Tall oil
2.4 Cutting
2.5 A nip

(5 × 1) **[5]**

QUESTION 3

3.1 C
3.2 H
3.3 A
3.4 G
3.5 D

(5 × 1) **[5]**

QUESTION 4

4.1 T'sai Lun (1)

4.2
- Sappi fine paper
- Sappi forest
- Sappi kraft
- Sappi saiccor

(Any 3 × 1) (3)

4.3 Landfills (1)

4.4
- Method of the original fibre preparation.
- Nature of any printing, lamination and glueing that the paper had undergone.
- The purpose for which the paper had been used originally. (3)

4.5 Contraries are items that form part of the bales of secondary fibre,✓ but that are not supposed to be there e.g. staples, stones, plastic film and so forth.✓ (2)

4.6 A – Outer bark
B – Inner bark
C – Cambium
D – Sapwood
E – Annual ring
F – Heartwood (6)

4.7 •Rain fall
•Length of nodes and internodes
•Variety and age of sugar cane
•Sand
•Pith content (Any 3 × 1) (3)
[19]

QUESTION 5

5.1 The method is carried out in a room with 50% humidity at temperature of 23 °C✓ using paper that has reached its equilibrium moisture.✓ The sheet is weighted at this point, and again at 105 °C.✓ The moisture content is then reported as a percentage of the weight of the sheet, in comparison to its weight at 50% humidity.✓ (4)

5.2 It is a lightweight paper product with absorbent properties. (1)

5.3
- Toilet tissue
- Kitchen towels
- Facial tissue
- Napkins
- Hand towels
- Wipes (Any 4 × 1) (4)

5.4 5.4.1
- Bark darkens the pulp
- Bark consumes chemicals
- Bark leaves dark spots in the final paper
- Bark in the pulp leads to paper with poor strength
- Bark contains relatively small amounts of useful fibre

(Any 4 × 1) (4)

5.4.2 Drum debarker (1)

5.5 •Rocks
•Tramp metal
•Plastics
•Sand and grit (4)
[18]

QUESTION 6

6.1 A – Wood magazine
B – Sharpening lathe
C – Burr
D – Pulpstone
E – Pit
F – Flume channel
G – Shower
H – Finger bars (8)

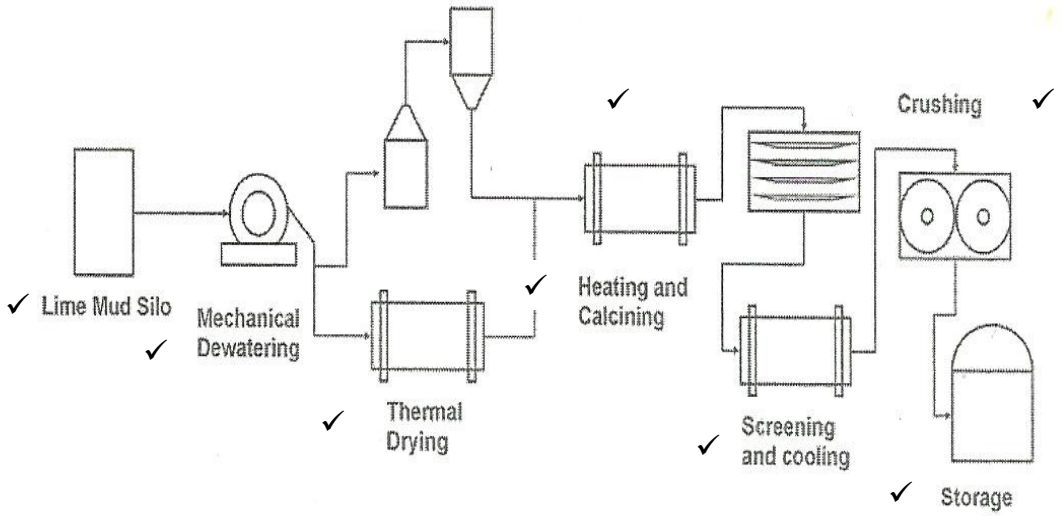
6.2 •NaOH (Sodium hydroxide)
•Na_2S (Sodium sulphide) (2)

6.4 •Batch digester
•Continuous digester (2)
[12]

QUESTION 7

7.1
- Avoid any excess air in the furnace.
- Keep the furnace temperature as high as possible.
- Use refractory lined piping because iron is a catalyst for the formation of SO_3.
- Cool the gas very rapidly because SO_3 forms between 600 and 900 °C. (4)

7.2



MARKS FOR CORRECT LABELLING AND 1 MARK FOR CORRECT DIAGRAM (8)
[12]

QUESTION 8

8.1
- Good formation
- Less linting of sheet
- Two-sidedness much improved
- Increased drainage capacity/speed increase
- Fewer stationary vacuum units, flat boxes or wet suction boxes
- Improve paper quality
- Affordable installed cost for retrofits (Any 5 × 1) (5)

8.2 8.2.1 The total pressure of the sheet starts to grow through compression.✓ Air flow out of both paper and felt.✓ No hydraulic pressure is present.✓ Felt and paper become saturated.✓ Transfer of water can only occur through capillary forces or two-phase flow.✓ Very little change in dryness occurs. All forces are taken up by compression of the fibre structure.✓ (6)

8.2.2 It is the removal of water from a sheet✓ to consolidate the sheet.✓ (2)

[13]

QUESTION 9

9.1 9.1.1
- Weighing✓ – the roll mass is determined.✓
- Wrapping✓ – the roll is wrapped in a protective material.✓
- Labeling✓ – labels are applied to show reel number, mass, grade, size and length.✓ (3 × 2) (6)

9.1.2
- Embossing
- Supercalendering
- Laminating
- Coating (Any 2 × 1) (2)

9.2
- Taking rolls from mill stock and rewinding them to a specific size, diameter and core size to fill small order specification.
- Slitting narrower widths that cannot be slit on the winder.
- Rewinding to smaller diameter that cannot be wound on machine winder. (3)

[11]

TOTAL: 100

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